

Section-C

(Long Answer Type Questions)

Note : There are **four** questions in this section. Attempt **only one** question. Each question carries **15** marks. [1 × 15 = 15]

6. Given $\frac{dy}{dx} = \frac{y-x}{y+x}$, with $y = 1$ for $x = 0$. Find y for $x = .1$ by Euler's method in five steps.

7. Use Runge-Kutta method to solve $\frac{dy}{dx} = xy$ for $x = 1.4$. Initially $x = 1$, $y = 2$ (take $h = 0.2$).

8. Solve the Simultaneous equation by using Gauss-Seidel Method :

$$10x + y + z = 12$$

$$2x + 10y + z = 13$$

$$2x + 2y + 10z = 14$$

9. Find $f(x)$, where :

x	0	1	2	3	4	5
f(x)	0	3	8	15	24	35

-----XX-----

Roll No. :

BCA-5004

BCA (Semester-V) (NEP) Examination, 2025-26

NUMERICAL METHODS

Time : 2 Hours]

[Maximum Marks : 75

Note :1. Attempt questions from **all** sections as directed.

2. The candidates are required to answer in serial order only. If there are many parts of a question, answer them in continuation.

3. "B" Copy will not be provided.

Section-A

Short Answer Type Questions

Note : There are Nine questions in this section. Attempt all questions. Each question carries **5** marks. [9 × 5 = 45]

1. (a) Derive the relationship between the forward difference operator and the backward difference operator.

- (b) Find the function whose first difference is :

$$3x^2 + 5x + 7$$

- (c) Obtain the missing term in the following table :

x	0	1	2	3	4	5	6
y = f(x)	1	2	?	8	16	?	64

- (d) Using Lagrange's Interpolation, find f(4) from the following table :

x	0	1	2	5
f(x)	2	5	7	8

- (e) Find the first derivate of the function f(x) at the point x = 1.5 if :

x	1.5	2.0	2.5	3.0	3.5	4.0
f(x)	3.375	7.000	13.625	24.000	38.875	59.000

- (f) Apply Newton Backward formula to find population in year 2017 from the following data :

Year	Population (In Thousands)
2010	45
2012	55
2014	68
2016	83

- (g) Write the steps involved in solving a system of linear equations using the Gauss Elimination Method.

- (h) Calculate the value of $\int_{-3}^3 x^4 dx$ by Simpson's $\frac{1}{3}$ rule and Trapezoidal rule.

- (i) Write the steps of Bisection Method.

Section-B

Long Answer Type Questions

Note : There are **four** questions in this section. Attempt **only one** question. Each question carries **15** marks. [1 × 15 = 15]

- Explain in brief about Falsi method. Solve $x^3 - 9x + 1 = 0$ for the root lying between 2 and 4 by Falsi method.
- Use Stirling's formula to find y_{35} , given :
 $y_{20} = 512, y_{30} = 439, y_{40} = 346, y_{50} = 243$
- Derive the Newton's Divided difference formula.
- Derive the Newton-Raphson method from the Taylor's series. Use Newton-Raphson method to find a root of $f(x) = x^3 - x - 1 = 0$ taking $x_0 = 1$, correct up to three decimal places.